Printed Pages : 4

Semester : V

TCS05027

Roll No. 2108L22037

Computer Science and Engineering

December-2023

DESIGN & ANALYSIS OF ALGORITHM

Maximum Marks : 70

Time Allowed : 3 Hours

Note: i) Attempt all the questions

ii) Maximum marks assigned to each question is indicated.

Section - A

Q.1. Attempt all questions (2 marks questions). **(2x5=10)**

(a) Calcualte time complexity for the given pseudocode. (BT-4, CO-1, PO-1)

```
for (i=1; i<n; i=i*2)
{statements;
}
```

(b) Explain min-heap ordering property. (BT-1, CO-2, PO-1)

(c) State the principle of optimality. (BT-1, CO-3, PO-1)

(d) Define chromatic number(BT-2, CO-4, PO-1)

(e) Discuss time complexity of naive string match algorithm. (BT-1, CO-5, PO-2)

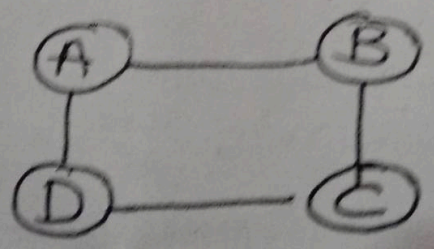
Attempt all questions (short answer questions). (4x5=20)

(f) Write a brief description on Asymptotic notation. (BT-2, CO-1, PO-1)

(g) State the properties of a Red Black tree and also calculate its time complexity. (BT-3, CO-2, PO-1)

(h) Differentiate between dynamic programming and greedy approach. (BT-2, CO-3, PO-2)

(i) Explain Graph coloring Problem. Determine the number of colors used for the graph given below. (BT-1, CO-4, PO-1)



(j) Explain Knuth-Morris Pratt algorithm. Draw π table for the pattern ababd. (BT-3, CO-5, PO-3)

(c) State the principle of optimality. (BT-1, CO-3, PO-1)

(d) Define chromatic number(BT-2, CO-4, PO-1)

(e) Discuss time complexity of naive string match algorithm. (BT-1, CO-5, PO-2)

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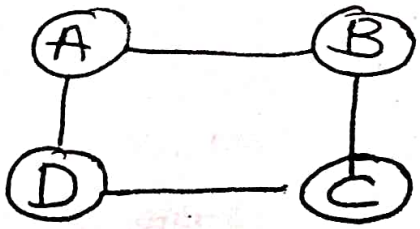


(j) Explain Knuth-Morris Pratt algorithm. Draw π table for the pattern ababd. (BT-3, CO-5, PO-3)

Section - B

Q.2. Calculate the time complexity for the given recurrence relation using substitution method. 8

$T(n) = T(n-1) + \log n$ (BT-4, CO-1, PO-3)

OR

Explain merge sort algorithm and trace this algorithm for n = 8 elements. (BT-2, CO-1, PO-1)

{24, 12, 35, 23, 45, 34, 20, 48}

Q.3 Let A {7, 2, 4, 17, 1, 11, 6, 8, 15, 10, 20}. Draw binomial heap for above elements.8 (BT-1, CO-2, PO-2)

OR

Explain B-Tree and construct a B-Tree for following elements. 8(BT-1, CO-3, PO-2)

{5, 3, 21, 9, 1, 13, 2, 7, 10, 12, 4, 8}

Q.4. Find an optimal sequence to the n = 5 jobs where profits $(P_1, P_2, P_3, P_4, P_5) = (20, 15, 10, 5, 1)$ and deadlines (BT-2, CO-3, PO-3)

$(d_1, d_2, d_3, d_4, d_5) = (2, 2, 1, 3, 3)$

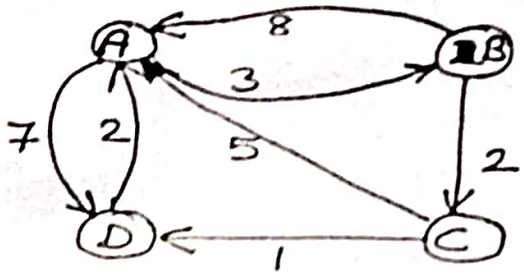
OR

Explain Longest common subsequence. Find the longest common subsequence for X = <A, B, C, B, D, A, B> and Y =<B, D, C, A, B, A>

(BT-2, CO-3, PO-3)

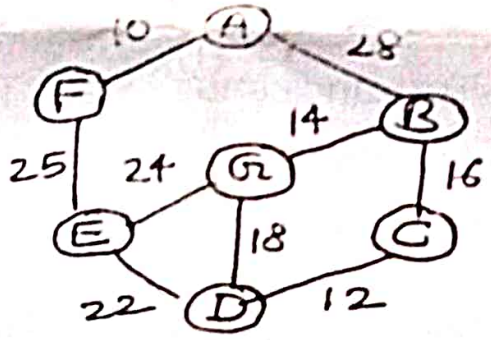
Q.5. Explain Floyd-Warshall Algorithm. Determine the shortest path for the given graph. 8

(BT-4, CO-4, PO-2)



OR

Explain a spanning tree. find the minimum cost of the given graph using Prim's algorithm. (BT-4, CO-4, PO-3)



Q.6. Explain Rabin Karp string matching algorithm. Also determine for q=11, how many valid and spurious hit found for given text and pattern P=26.

(BT-2, CO-5, PO-1)

T = 3 1 4 1 5 9 2 6 5 3 5 8 9 7 9 3

OR

Explain NPhard and NP complete in detail.

(BT-1, CO-5, PO-2)

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TCS05027

Semester : V

Roll No. .....

B.Tech.

Examination, January-2023

(DESIGN & ANALYSIS OF ALGORITHM)

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Time Allowed : 3 Hours

Note: i) Attempt all the questions.

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Section - A

Q.1. Attempt all questions (2 marks questions) (2x5=10)

(a) Explain five main characteristics of an algorithm

(b) Discuss different types of rotations in RB trees.

(c) Differentiate between dynamic programming and greedy algorithm.

(d) Illustrate Depth First search with suitable example

(e) Write short note on NP-Complete problems.

Attemp all questions (Short answer questions)

(4x5=20)

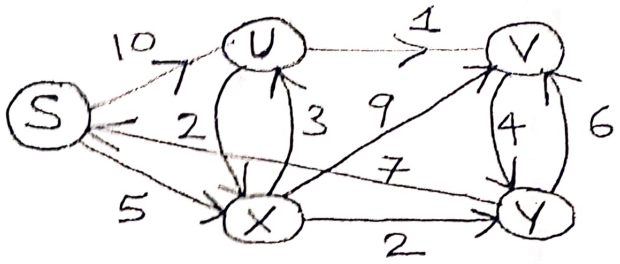
(f) Illustrate the operation of HEAP SORT on following array A = {6, 14, 3, 25, 2, 10, 20, 7, 6}

(g) Draw a binomial heap whose keys are element of A

A = {7, 2, 4, 17, 1, 11, 6, 8, 15, 10, 20}

(h) Explain in detail about 0/1 knapsack problem and fractional knapsock problem

(i) Find the shortest path in the following directed graph from the source vertex S to Y by using dijikstra algorithm



(j) Define naive string matching algorithm with suitable example.

Section - B

Q.2. Discuss the brief: best case, average case and worst case time complexity of analgorithm 8

OR

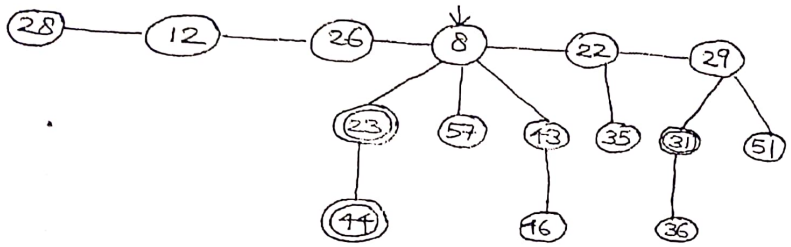
Solve the following recurrence relation

a) $T(n) = T(a,n) + T(1-a) + n, 0<a<1$

b) $T(n) = T(an/10) + n$

Q.3 Consider a Fibonoccl heap shown in below figure. Run the Fib-Heap extract-min and give the

resultant Fibonaccl Heap here double circle represent marked mode.



OR

Show the result of inserting the following items in an initally empty is tree of order 3

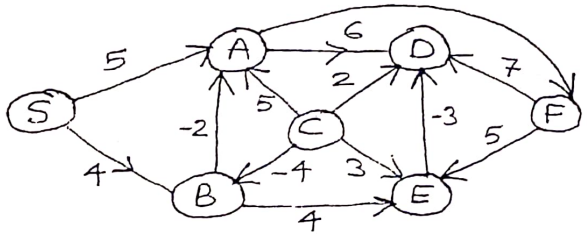
F, S, Q, K, C, L, H, T, V, W, M, R, N, P, A, B, X, Y D, Z, E, G, I

Q.4. Define Assembly Line scheduling problem and task scheduling problem. 8

OR

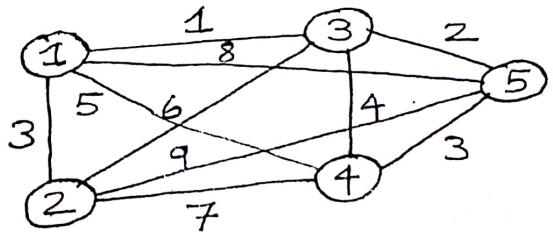
Discuss matrix chain multiplication with reference to dynamic programming technique whose sequence of dimension is <30, 1, 40, 10, 25>

Q.5. Find the shortest path using BELLMAN FORD algorithm form S to F in the following graph



OR

Solve the travelling salesman problem for the following graph by using branch and bound algorithm



8

Q.6. Write short note on:

a) Knuth morris pratt algorithm for pattern matching

b) Randomized Sorting Algorithm

OR

Discuss P, NP-Complete problems, NP and NP-hard problems using suitable example